

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 13

06-08-2026

13. Write the python program to implement Minimax algorithm for gaming.

Aim

To implement the Minimax Algorithm in Python for a two-player game. The objective is to determine the optimal move by maximizing the player's score while assuming that the opponent always plays optimally to minimize the score.

Algorithm

1. Start the program.
2. Define the terminal or leaf values of the game tree.
3. Construct the game tree with MAX and MIN levels.
4. At each MIN node, select the minimum value among its children.
5. At each MAX node, select the maximum value among its children.
6. Propagate the selected values towards the root.
7. Determine the final optimal value at the root MAX node.
8. Display the leaf values and optimal value.
9. Generate the game tree using NetworkX.
10. Display the game tree using Matplotlib.
11. Stop the program.

Program

```
import networkx as nx

import matplotlib.pyplot as plt

v = [3,5,2,9]

a = max(min(v[0],v[1]),min(v[2],v[3]))
```

```
print("Leaf Values:",v)
print("Optimal Value:",a)

G = nx.DiGraph()

G.add_edges_from([
    ("MAX","MIN1"),("MAX","MIN2"),
    ("MIN1","3"),("MIN1","5"),
    ("MIN2","2"),("MIN2","9")
])

p = {"MAX":(0,2),"MIN1":(-1,1),"MIN2":(1,1),
     "3":(-1.5,0),"5":(-.5,0),
     "2":(.5,0),"9":(1.5,0)}

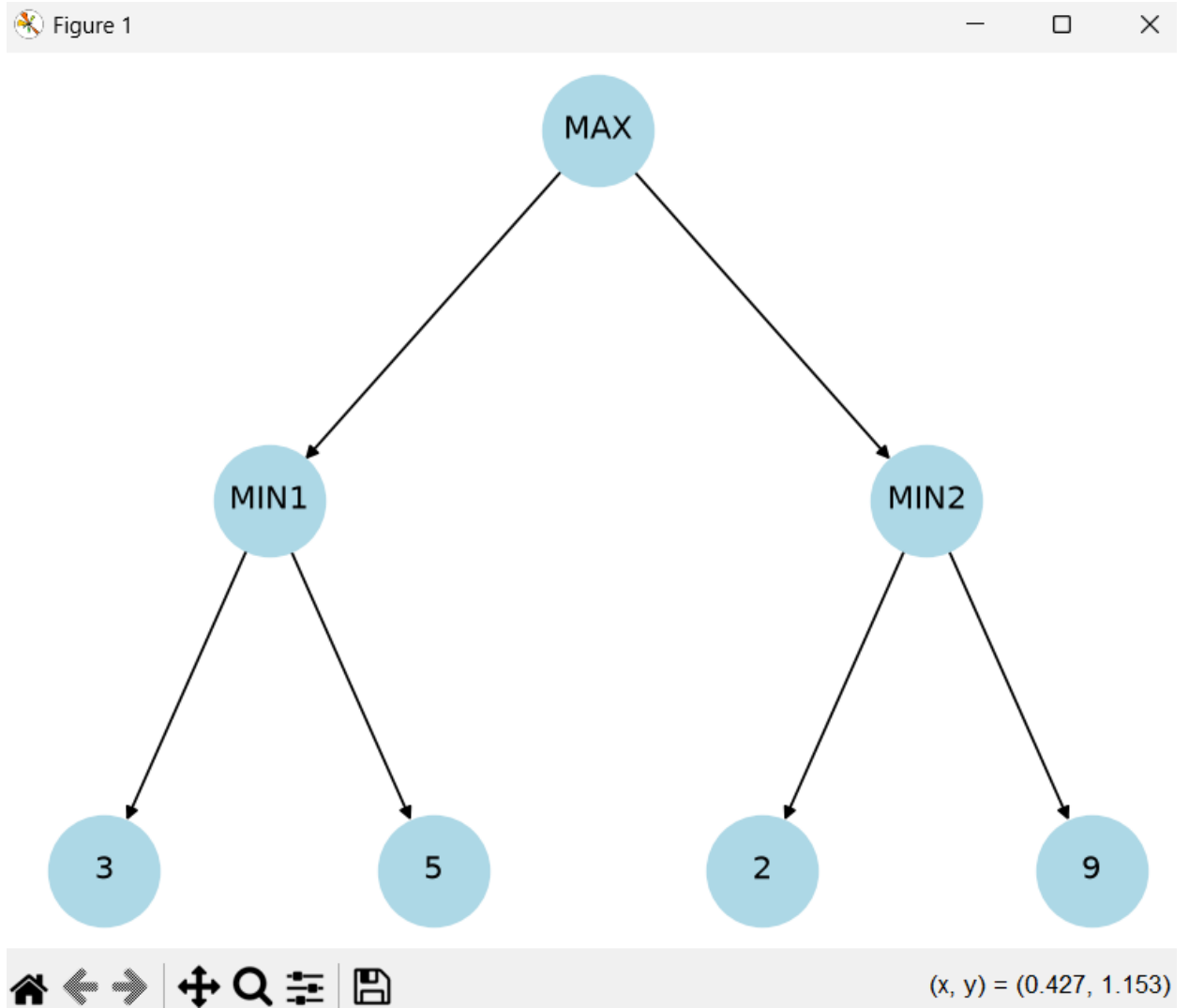
nx.draw(G,p,with_labels=True,node_size=1800,
        node_color="lightblue")

plt.title("Minimax Game Tree")
plt.show()
```

Output:

Leaf Values: [3, 5, 2, 9]

Optimal Value: 3



Result:

The Minimax Algorithm was implemented successfully in Python. The algorithm evaluated all possible game outcomes by alternating between MAX and MIN players and determined the optimal value for the maximizing player.